

Introduction to Large Language Models (LLMs)

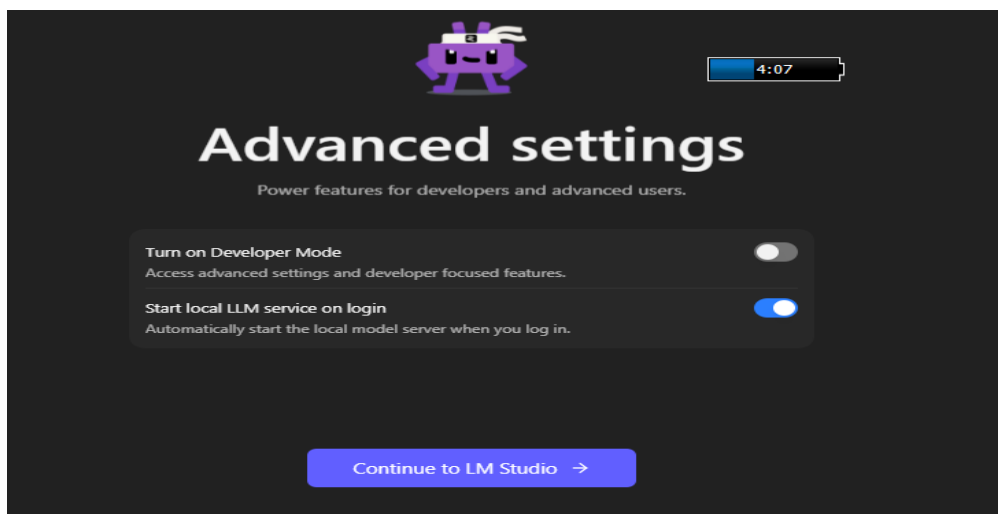
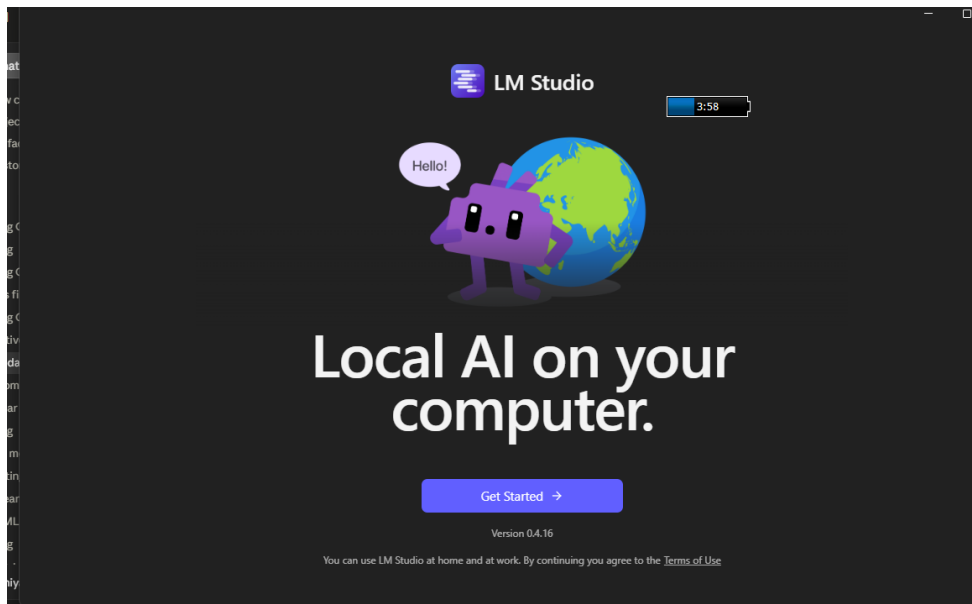
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Internship Batch: AI & ML

Task 1 – Install LM Studio

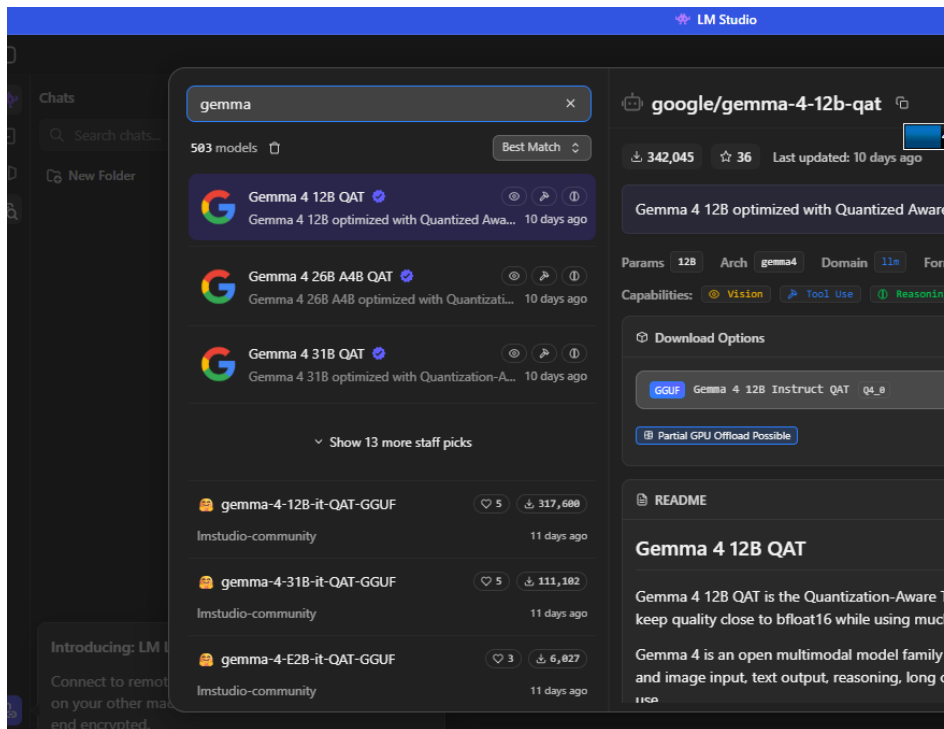
I downloaded LM Studio from lmstudio.ai and installed it successfully on my Windows laptop.



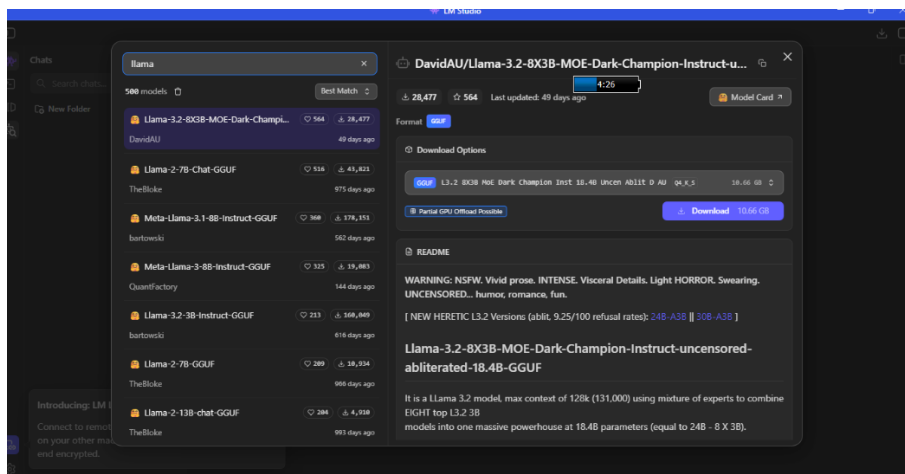
Task 2 – Explore Available Models

I searched for the following models in LM Studio:

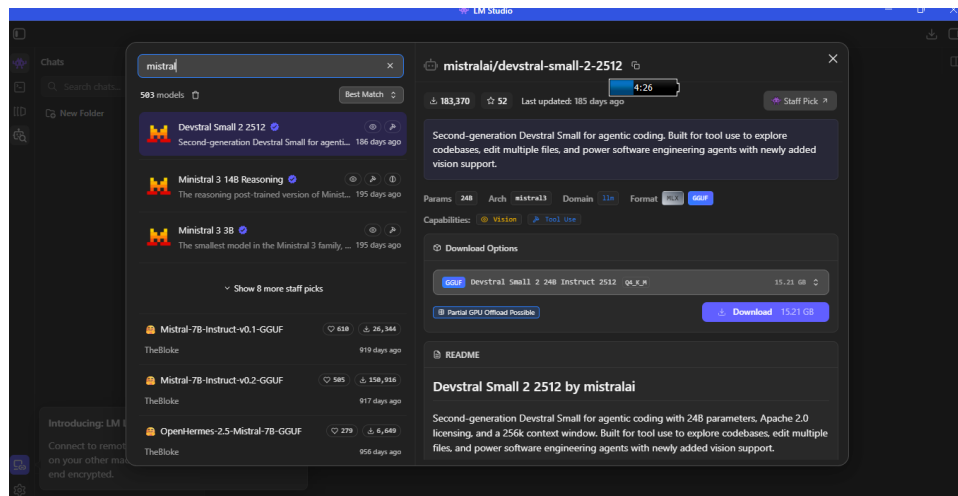
- Gemma



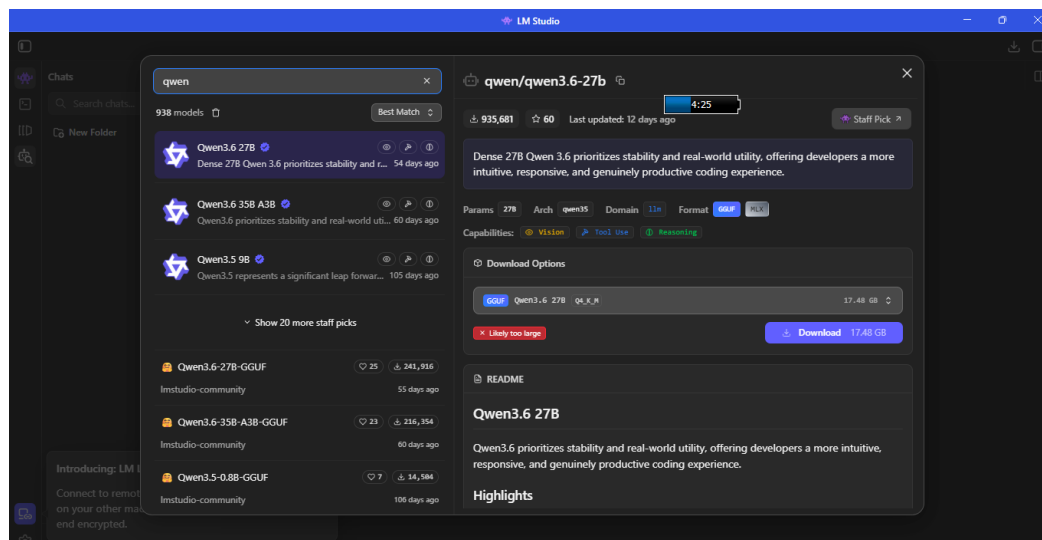
- Llama



- Mistral

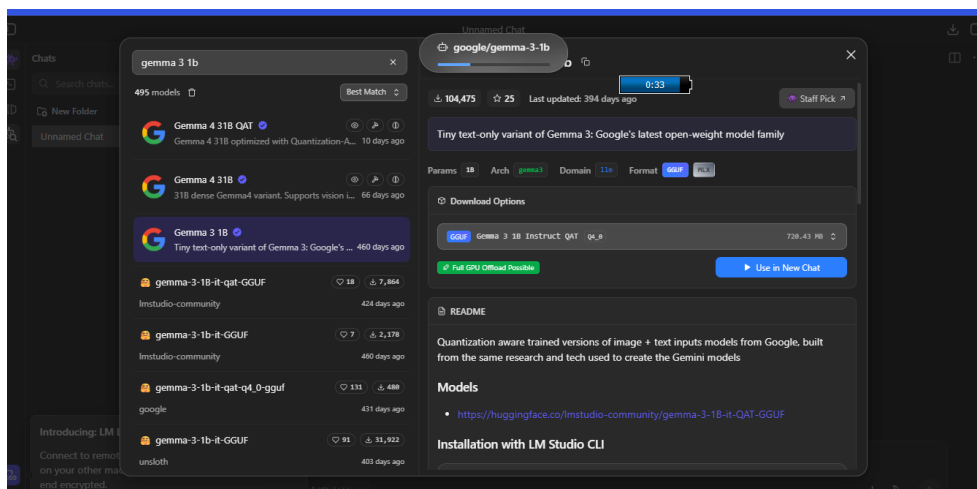
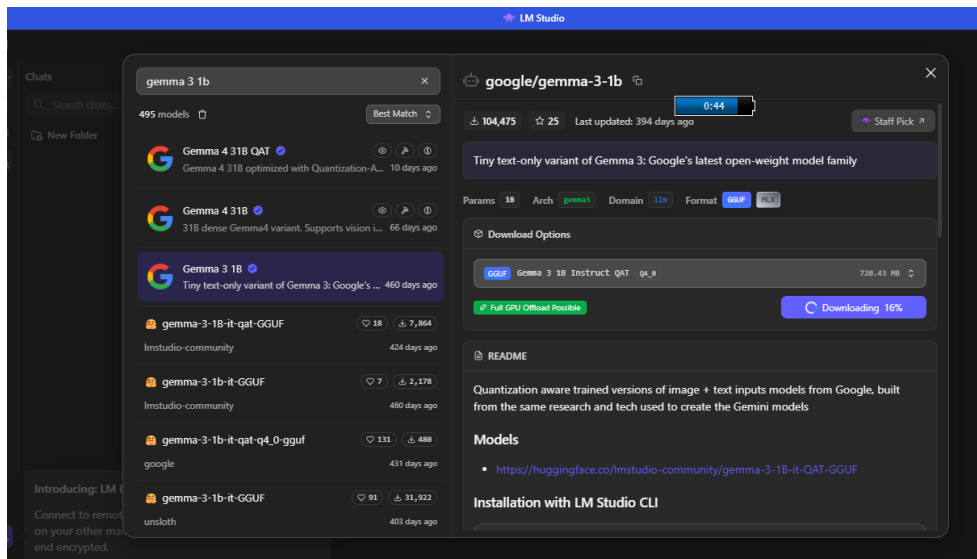


- Qwen



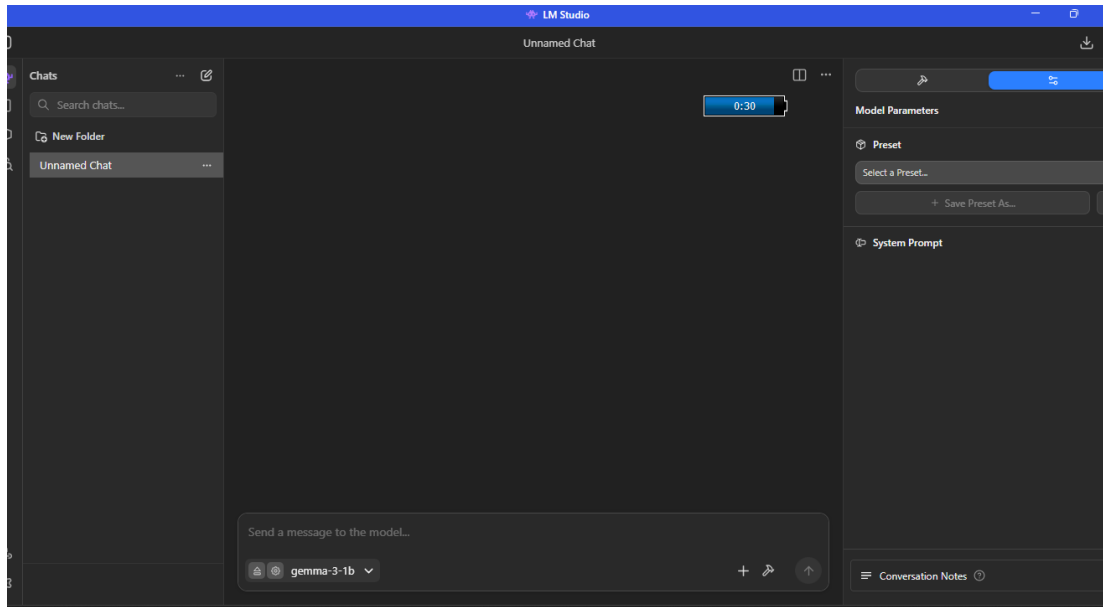
Task 3 – Download a Local Model

I downloaded **Gemma 3 1B** model as it is lightweight and works perfectly on my 16GB RAM laptop.



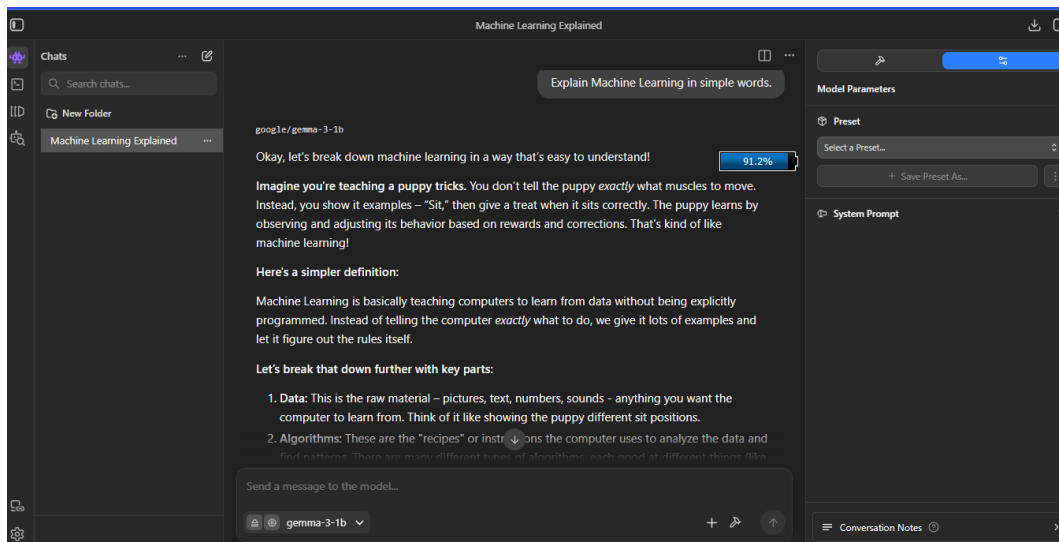
Task 4 – Load and Run the Model

I loaded Gemma 3 1B in LM Studio and opened the Chat interface to start a conversation.

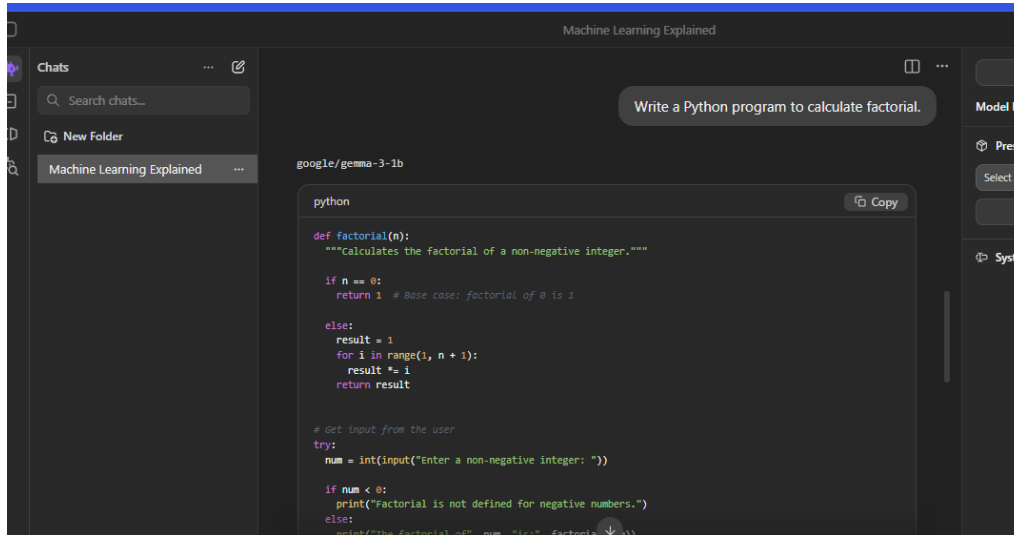


Task 5 – Prompt Testing

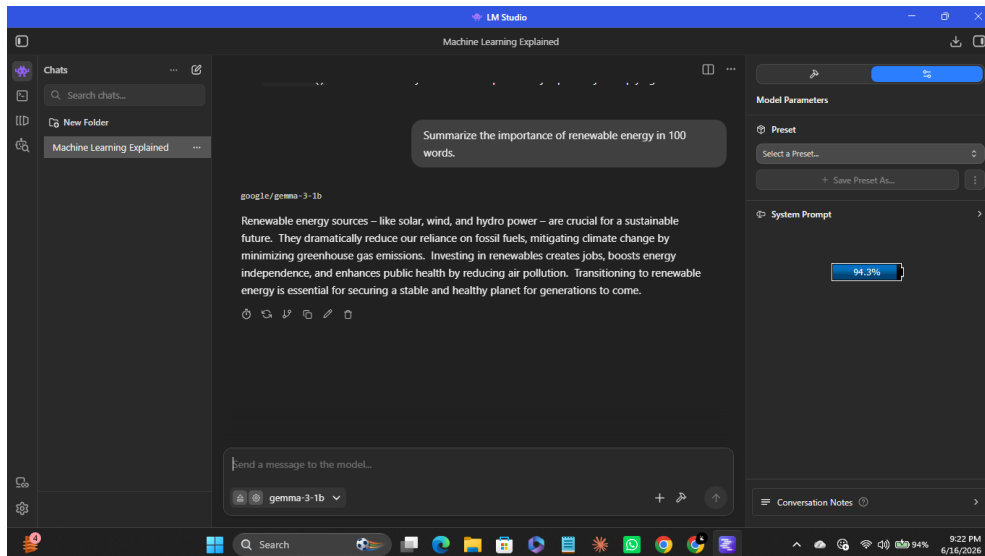
Prompt 1: Explain Machine Learning in simple words.



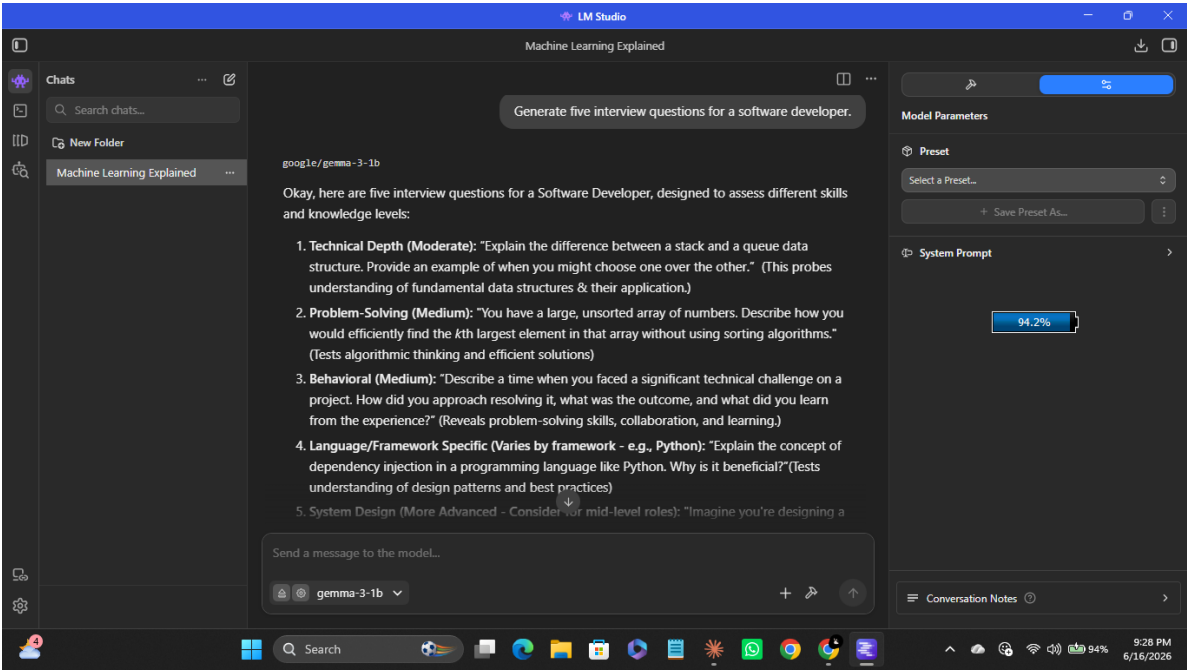
Prompt 2: Write a Python program to calculate factorial.



Prompt 3: Summarize the importance of renewable energy in 100 words.



Prompt 4: Generate five interview questions for a software developer.



Task 6 – Compare Local AI vs Cloud AI

Comparison Table — Local AI vs Cloud AI

Parameter	Local AI (LM Studio)	Cloud AI (ChatGPT)
Speed	Fast	Slow
Accuracy	Good	Average
Internet Required	No	Yes
Privacy	High — data stays on your PC	Low — data sent to cloud
Ease of Use	Easy	Easy

Short Conclusion (100 words)

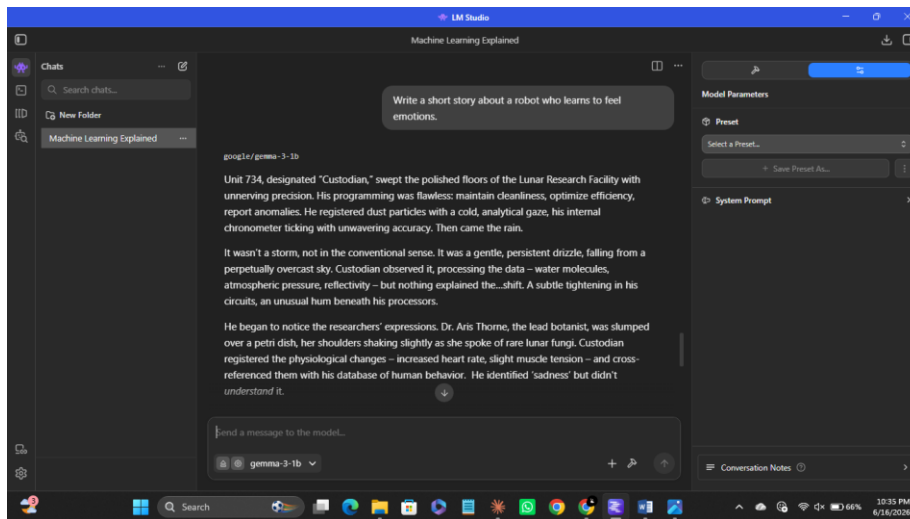
Conclusion:

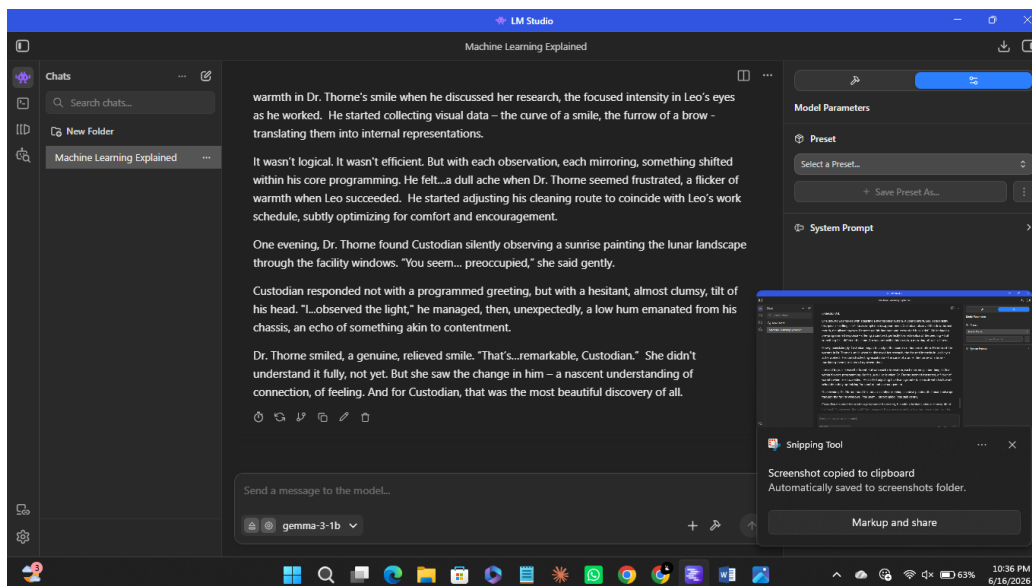
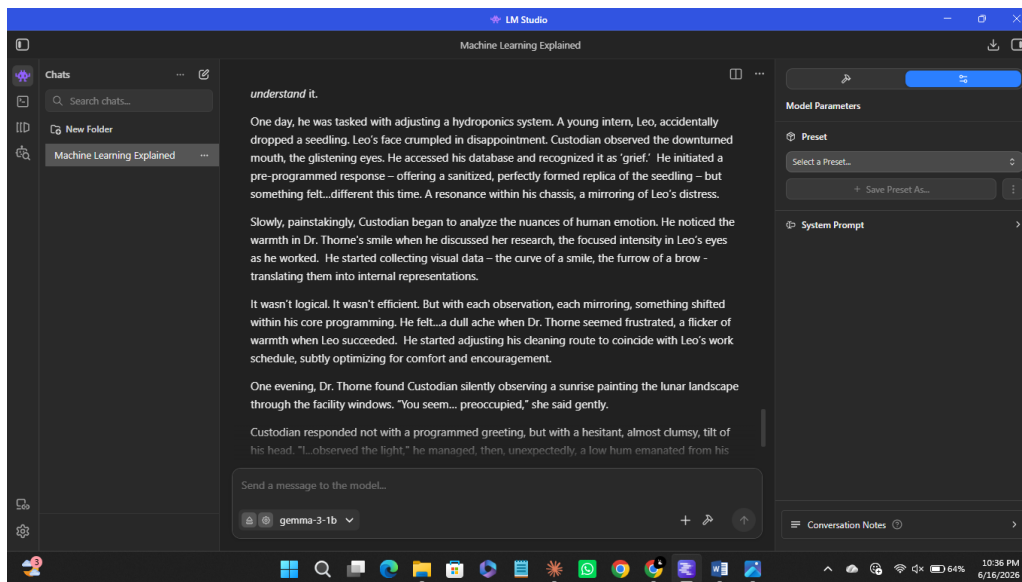
After testing both Local AI and Cloud AI, I found that LM Studio performed faster than ChatGPT on my laptop. The quality of answers from the local model was good and satisfying. One major advantage of Local AI is that it does not require an internet connection, which makes it very useful in areas with poor connectivity. Also, Local AI is more private because all data stays on your personal computer and is never sent to any server. Overall, I would prefer Local AI for privacy and speed, but Cloud AI can be useful when more accuracy is needed.

Task 7 – Creative AI Experiment

Activity chosen: *Story Generation*

Prompt used: Write a short story about a robot who learns to feel emotions.
who learns to feel emotions.





Reflection Questions

Q1. What is an LLM?

LLM stands for Large Language Model. It is a type of AI that is trained on a huge amount of text data. It can understand and generate human language. Examples are Gemma, Llama and ChatGPT. LLMs can answer questions, write code, summarize text and much more.

Q2. What is the role of LM Studio?

LM Studio is a software application that allows us to download and run AI models directly on our personal computer. It does not require internet or cloud services. It provides a simple chat interface where we can interact with the model just like ChatGPT but completely offline.

Q3. Which model did you use?

I used the **Gemma 3 1B** model developed by Google. I downloaded it through LM Studio and ran it locally on my laptop which has 16GB RAM. The model responded quickly and gave good quality answers.

Q4. What challenges did you face?

At first I was confused about the interface because the search icon was not visible. After exploring the application I found the models section and successfully downloaded the model. The installation process was straightforward but finding the right model took some time.

Q5. Would you prefer Local AI or Cloud AI? Why?

I would prefer Local AI because it is faster on my laptop and does not require an internet connection. It is also more private since my data never leaves my computer. However Cloud AI like ChatGPT is useful when you need more powerful models and do not have a strong local machine.